



Society of Petroleum Engineers

SPE-229795-MS

Constructing Mega Offshore: Validating Numerical Data Through Scaled Physical Simulation

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229795-MS](https://doi.org/10.2118/229795-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

This paper presents a comprehensive procedure for conducting a float-over model test for a mega topside installation, focusing on validating numerical predictions with physical test results. It outlines the entire testing process, including equipment, data collection, and post-processing methods, and compares the outcomes with analytical simulations performed using MOSES software. A 1:40 scale model was constructed in an Ocean Engineering Basin at Shanghai Jiao Tong University in China. The basin, measuring 50m in length, 30m in width, and 6m in depth, was used to simulate environmental conditions such as waves, currents, and winds. More than 200 test scenarios were performed covering all floatover stages from docking, entrance, zero clearance, 0% load transfer up to undocking. The float-over model testing involved model scaling, construction, calibration, system response tests, and float-over simulations in irregular waves. Calibration was a critical step to ensure the accuracy of measurement instruments and data acquisition systems. Raw data from all tests, including calibration, were recorded and later statistically processed to derive motion, acceleration, and loading responses. Special attention was given to identifying and eliminating spurious data, as such anomalies could significantly affect the test results. The methodology for filtering these spurious values was crucial to improving the reliability of the output. The final model test results were then benchmarked against analytical outputs from MOSES simulations. The comparison between the float-over model tests and numerical simulations demonstrated strong agreement, reinforcing the accuracy and reliability of the analytical predictions. These findings validate the use of MOSES software for such installations and were instrumental in critical design decisions, such as the selection of jacket fenders and LMU (Leg Mating Unit) features. The diverse test scenarios facilitated detailed observations of the multi-body system behavior across all critical stages of the float-over installation of the mega topsides, which was essential for validating the numerical simulation results and enhancing the reliability of the design.

Introduction

The new Marjan GOSP-4 (Gas-Oil Separation Plant) facility is a mega offshore development designed to meet the increased production capacity requirements of the Marjan Field. This facility comprises six (6) mega platforms namely Tie-in (TP), Production (PP), Gas Compression (GCP), Auxiliary (AUX), Accommodation (ACC), and Electrical Distribution (EDP) along with a flare structure, all interconnected by eleven (11) bridges. The topsides of these platforms have installation weights ranging from 11,000 MT to 23,700 MT (excluding DSF weight). All platforms were successfully installed using the floatover method at a water depth of 52 meters.

Among the six (6) topsides, the Tie-in (TP) and Production (PP) topsides are the heaviest, weighing approximately 20,600 MT and 23,700 MT, respectively. Although the PP topside is heavier, it was transported and installed using a larger Heavy Transport Vessel (HTV) with a deadweight of approximately 66,000 MT. In contrast, the TP topside was installed using a smaller HTV with a deadweight of approximately 48,500 MT. This configuration introduced greater technical challenges and higher operational risks for the TP floatover operation. Additionally, the floatover direction for the TP topside was perpendicular to the predominant wave direction, further increasing the complexity of the installation. Due to these factors, the TP topside floatover was selected for the physical model test to better assess and mitigate the risks associated with floatover hardware design and the actual installation operations.

Despite advancements in numerical simulations, their accuracy is constrained by idealized assumptions that often fail to capture the complex multi-body interactions during floatover operations. These limitations can lead to inaccuracies in predicting vessel motion responses, and impact loads on LMUs, DSUs, and fender systems. For high-risk installations like the TP floatover, physical wave basin model testing is essential to validate engineering designs, evaluate critical floatover stages, and optimize floatover hardware design. Integrating numerical simulations with physical model test data ensures a more robust and reliable floatover operation planning.

The floatover installation requires integration of multiple systems, including DP control on the semi-submersible vessel, jacket, topside, fenders, DSF, DSU, and LMU. Advanced hydrodynamic tools like OrcaFlex, Sima, and MOSES allow comprehensive time domain simulations of the entire operation. However, validating these simulations through scaled physical model testing remains a significant challenge. Li et al. [1] reported the ninth global and first Chinese DP floatover project (HZ-258DPP), featuring the first model-scale tests for DP-controlled docking and undocking. He et al. [2] detailed these tests and developed a time-domain simulation program that aligned well with physical results. Wang et al. [3] analyzed a twin-barge floatover system with flexible connections and DP control through combined numerical and model testing. Wang et al. [4] introduced a feed-forward DP control strategy for swell conditions, validated through model tests, which effectively reduced thruster wear.

Unlike previous studies, this paper presents a model-scaled physical simulation of mega floatover operations for the Marjan Increment Project. Experimental data from docking, mating, and undocking tests are compared with MOSES-based numerical simulations. The setup, data acquisition, and post-processing methods are detailed in subsequent sections, along with the DP control approach used.

Experimental Set-ups

A 1:40 scale physical simulation was conducted in the Ocean Engineering Basin at Shanghai Jiao Tong University (Xuhui Campus) to validate analytical predictions of system responses, including dynamic motions, impact forces, and the DP control system used in floatover operations. The basin, measuring 50 m × 30 m × 6 m (see Fig.1), enables realistic ocean condition simulations with adjustable water depth. It features a multi-flap wave generator for regular and irregular waves, axial wind fans (up to 10 m/s), and a high-pressure jet system where current is controlled by adjusting the rotational speed of individual pumps. An absorbing beach minimizes wave reflections, ensuring accurate wave modeling. The facility

is equipped with 6-DOF optical motion tracking, force and acceleration sensors, wave height meters, and a dynamic positioning test system to evaluate station-keeping under combined environmental loads. The experimental setup, data acquisition, and post-processing procedures ensured consistency and accuracy. These capabilities enabled comprehensive validation of numerical simulations performed using MOSES, particularly for docking, mating, and undocking stages.

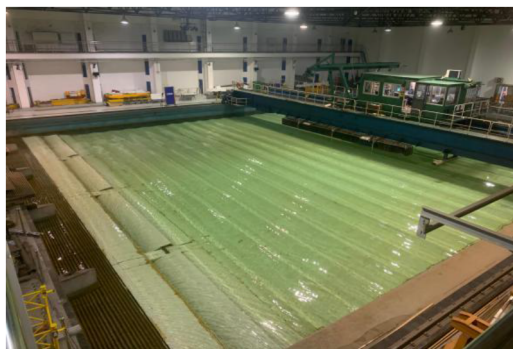


Figure 1—Ocean Engineering Basin, Shanghai Jiao Tong University

Model Preparation

The floatover model test program involved the construction of the vessel, thruster units, jacket, topside, LMUs, DSUs, and fender systems, all fabricated in accordance with engineering designs and floatover installation requirements. The X-class vessel model was built using wood and sealed with watertight materials such as glass fiber reinforced plastic. Internal compartments were allocated for ballast to adjust draft, trim, and mass distribution, ensuring the hull form, hydrostatic properties, and inertia characteristics remained within 1% of the full-scale values (see Fig.2 and Table-1). The propulsion system comprised two bow thrusters, two stern thrusters, and two main thrusters, each fabricated according to force similarity theory to accurately replicate maneuvering behavior during standby, docking and undocking stages (see Fig. 3 and 4). The vessel deck was reinforced to support the topside load during mating. The steel jacket model represented the full water depth and was fixed to the basin floor to ensure stability during docking and undocking phases. The topside model was constructed with appropriate mass, stiffness, center of gravity, and wind area to reflect realistic floatover conditions (See Fig.5). Eight LMUs were installed between the topside and jacket legs, each comprising vertical and horizontal spring elements and interfaced with stabbing cones to guide and align the topside while avoiding steel-to-steel contact (see Fig.6). DSUs mounted on the vessel deck supported the topside during float-in and load transfer, incorporating friction materials to manage impact loads and restrict motion. Sway fenders were fabricated using a geometry-similar contact face and a stiffness-similar spring and installed on each jacket leg with a nominal 2.5 mm clearance from the vessel side shell to limit lateral movement, while surge fenders assisted in achieving the correct pre-mating position (see Fig.7). All spring elements were designed to match the required force-deflection behavior, and local coordinate systems were defined for LMUs, DSUs, and fenders to ensure consistent force and motion representation throughout the test program.

Table 1—Main parameters of the vessel

Parameters	Unit	Full scale
Length over all	m	231.1
Moulded breadth	m	46
Moulded depth	m	14.5
Lightship draft	m	10.9
Scantling draft	m	27.6
Lightship weight	t	26800



Figure 2—X-Class vessel model



Figure 3—The thrusters and rudders model:(a) main thruster;(b) bow thruster

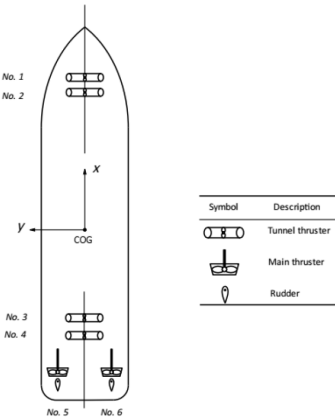


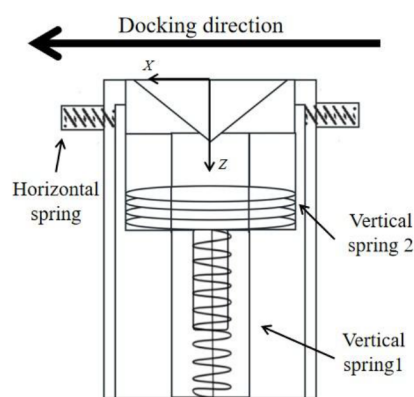
Figure 4—The configuration of the thruster system



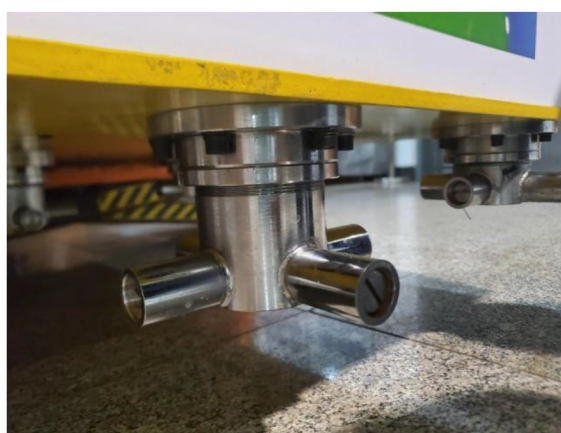
(a)

(b)

Figure 5—The steel model:(a) TP jacket;(b) TP topside



(a)



(b)

Figure 6—The design of the LMUs and DSUs:(a) The schematic of the LMU and DSU models; (b) The LMU and DSU models

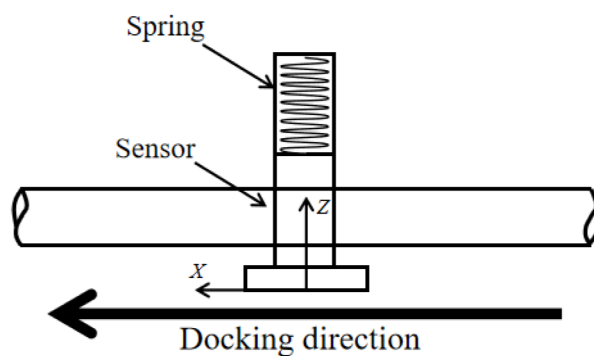


Figure 7—Experimental setup of fenders

Simulation of environmental conditions

Environmental condition simulation in floatover model testing as summarized in Table 3 is critical for ensuring dynamic similarity and accurate representation of offshore installation scenarios. The target water depth of 52.5 m is scaled to 1.312 m in the ocean engineering basin using a large-scale adjustable false bottom. This setup allows precise control of hydrostatic and hydrodynamic conditions, enabling realistic modeling of vessel and jacket interactions. The scaled depth ensures that low-frequency motions such as surge, sway, and yaw are properly captured during mating operations.

Table 2—Environment Conditions adopted in Model Test

Wave Direction	Wave Height H_s (m)	Wave Peak Spectral Period, T_p (s)	Wave Peakedness Parameter, γ	1hr Average Wind Speed, V_w (m/s)	Current Speed, V_c (m/s)
Stern 0°	1.5	4.5	4.6	12	0.5
	1.5	5.5	1.8	12	0.5
	1.5	7	1	12	0.5
Quartering 45°	1	4	3.16	10	0.5
	1	5	1	10	0.5
	1	6	1	10	0.5
Beam 90°	0.5	4	1	8	0.5
	0.5	4.5	1	8	0.5
	0.5	5	1	8	0.5

Table 3—Calibration Results for Wind Loads

Wind Direction	Wind Speed (m/s)	Fx		Error %	Fy		Error %	Mz		Error %
		Target	Measured		Target	Measured		Target	Measured	
0°	8	-0.25	-0.2545	1.8						
	10	-0.39	-0.3849	1.3						
	12	-0.56	-0.5769	3						
45°	8	-0.18	-0.1734	3.6	-0.33	-0.3404	3.1	-0.82	-0.754	-8.3
	10	-0.28	-0.2674	4.5	-0.51	-0.5297	3.9	-1.284	-1.318	2.6
	12	-0.4	-0.4063	1.6	-0.73	-0.7602	4.1	-1.85	-1.700	-8.1
90°	8				-0.46	-0.4805	2.1	-1.16	-1.207	4.1
	10				-0.72	-0.6941	3.6	-1.82	-1.746	-4.1
	12				-1.03	-0.9956	3.3	-2.59	-2.530	-2.3

Wind effects are reproduced using a computer-controlled wind generation system with fans and frequency converters. Wind speed at the model location is measured using a hot-wire anemometer, and wind loads are calibrated with the model fixed at a 3 m offset from the fan array. The calibration ensures that the one-hour mean forces and moments in surge, sway, and yaw match the numerical predictions as presented in Table 3 with mean wind force and moment within $\pm 5\%$ and $\pm 10\%$ tolerances, respectively. This process is essential for validating aerodynamic loading and its influence on vessel alignment and station-keeping during floatover.

Current simulation is achieved through a deep-water current generation system that circulates water to replicate realistic ocean current profiles. By adjusting the speed of high-pressure pumps, various vertical velocity profiles can be generated. Surface current velocities are measured using a Doppler current meter. Currents are applied in head-on, quartering, and beam directions during all irregular wave tests, allowing comprehensive evaluation of current-induced drift and its interaction with wave and wind forces.

Wave conditions include white noise waves and irregular waves based on the JONSWAP spectrum. Each irregular wave test spans three hours in prototype scale to capture sufficient cycles at low natural frequencies. Wave calibration is conducted using five wave probes positioned at the floatover location, with measurements taken in the absence of the model. Calibration results confirm that significant wave height

(Hs), peak spectral period (Tp), and spectral shape are within 5% and 3% of theoretical values, ensuring high accuracy in sea state reproduction (see Table 4).

Table 4—Main parameters of the vessel

Wave Direction	label No.	Wave Height, Hs(m)	Wave Peak Spectral Period, Tp(s)	Wave Peakedness Parameter, γ	1hr Average Wind Speed, Vw(m/s)	Current Speed, Vc(m/s)	Error of Hs	Error of Tp	Error of γ
WHN	0	2	4-20	-	-	-	-	-	-
Stern 0°	1	1.5	4.5		12	0.5	2.60%	-1.33%	0.04%
	2	1.5	5.5	1.8	12	0.5	0.40%	0.51%	3.83%
	3	1.5	7	1	12	0.5	2.67%	0.19%	0.00%
Quartering 45°	4	1	4	3.16	10	0.5	2.10%	0.92%	4.27%
	5	1	5	1	10	0.5	4.30%	1.74%	2.20%
	6	1	6	1	10	0.5	1.00%	0.17%	0.00%
Beam 90°	7	0.5	4	1	8	0.5	1.02%	0.05%	0.00%
	8	0.5	4.5	1	8	0.5	2.50%	1.47%	0.00%

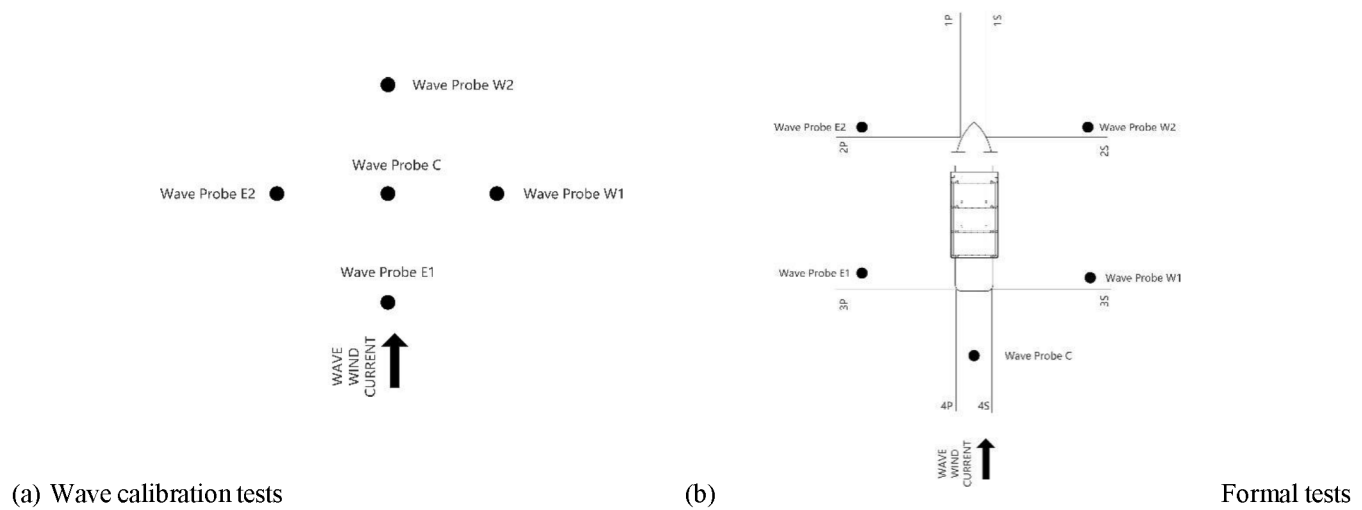


Figure 8—Configuration of the wave probes

Validation of linear transfer functions is performed using a long-crested irregular wave characterized by white noise, with an Hs of 2 m and a duration of three hours. This wave condition provides a broad frequency bandwidth suitable for evaluating vessel response across the full range of sea states. The environmental matrix includes combinations of wave direction, height, wind speed, and current speed, ensuring that the model test results are representative of realistic offshore conditions and suitable for engineering assessment of floatover operations.

Experimental procedure

The model test program commenced with a series of calibration and verification procedures to ensure the accuracy and reliability of the setup. Weight distribution calibration was performed by first determining the total weight and the center of gravity (CoG) of the empty model using bi-filar suspension. Ballast weights were then added to achieve the target weight and radii of gyration. The longitudinal and transverse GM values, as well as the CoG height, were verified through incline tests, while decay tests were used to confirm the natural periods in roll and pitch. Wave calibration was subsequently carried out by generating

a predefined wave spectrum in the basin without the model. The generated wave signals were analyzed and iteratively adjusted to meet the required spectral characteristics, including significant wave height (H_s), peak period (T_p), and spectral shape, within specified tolerances. Following this, white-noise wave tests were conducted to evaluate the vessel's response amplitude operators (RAOs) across a broad frequency range. These tests are essential for identifying the dynamic characteristics of the model and validating its hydrodynamic behavior under broadband wave excitation.

Following the initial calibrations, dynamic positioning (DP) system verification was carried out through DP standby tests in irregular waves. The vessel was first held in position using zeroing ropes in calm water, and all measurement signals were zeroed relative to this reference. After initiating the current pumps and achieving a steady flow, the DP system was activated, and the zeroing ropes were released. Once the vessel maintained its heading and position autonomously, wind and wave conditions were introduced sequentially. A single wave cycle was generated to eliminate transient effects, followed by a continuous three-hour (full-scale) sea state during which all relevant data and video recordings were collected. These procedures validated the DP system's capability to maintain station-keeping under combined environmental loads.

The floatover operation was simulated through a series of DP docking, mating, and undocking tests. The DP docking and undocking tests followed a similar procedure to the DP standby tests, with the vessel starting either 5 meters away from or within the jacket slot, respectively. The vessel was dynamically maneuvered using the DP system to complete the docking or undocking trajectory under irregular wave conditions. In contrast, the mating tests employed horizontal mooring lines in place of the DP system to simulate the surge control mechanism during the actual floatover operation. This setup was intended to represent the effect of the main thrusters maintaining continuous forward thrust to keep the surge fenders in compression during the mating phase. Pretensions in the mooring lines were adjusted accordingly, and the vessel was held in position in calm water. After zeroing the measurement signals, current, wind, and wave conditions were sequentially introduced. A single wave cycle was initially generated to allow initial transients to dissipate, followed by a continuous three-hour sea state during which load transfer and vessel responses were recorded. Over 200 model test scenarios were conducted, covering docking, mating, and undocking operations under various wave periods and directions including stern, quartering, and beam seas with multiple realizations for each condition.

Data Processing Procedure

The data processing was carried out in four main steps to ensure the quality and reliability of the experimental results. First, the initial 90 seconds of raw data including vessel motions, accelerations, loads, and wave heights were discarded to allow environmental loads to stabilize and meet the intended experimental conditions. This initial period was determined based on observed system behavior during the tests. Second, all data channels were manually inspected for spurious or anomalous values, which were removed to prevent errors in subsequent analysis. Third, high-frequency noise, primarily attributed to environmental white noise, was filtered out across all channels. This step also included the removal of unexpected load contributions arising from coupling effects and nonlinear motions, which could otherwise affect the accuracy of the system response. In the final step, static loads were incorporated into the dataset, and geometric and dynamic scaling laws were applied to convert the measured model-scale responses into their corresponding full-scale values. Statistical parameters such as mean, standard deviation, and peak responses were then computed to characterize the dynamic behavior of the system. This procedure ensures the processed data accurately represents the physical behavior observed during the floatover and provides a sound basis for evaluating the model test outcome.

DP controller design in physical simulation

The control objective is to keep the vessel's position and heading by compensating for only the low frequency motion, which is estimated from the observer. A fuzzy-Proportional-Integral-Derivative (PID) controller is used to track the predefined setpoint.

$$\tau_c = -\mathbf{K}_p \tilde{\eta}_p - \mathbf{K}_i \int_0^t \tilde{\eta}_p dt - \mathbf{K}_d \tilde{v} \quad (1)$$

Where $\tilde{\eta}_p = \mathbf{R}^T(\psi)(\eta - \eta_d)$ is the position and heading deviation vector in the body-fixed frame, $\tilde{v} = v - v_d$ is the velocity deviation vector, n_d and v_d are the desired position and heading and velocity vectors, respectively. $\mathbf{K}_p$, $\mathbf{K}_i$ and $\mathbf{K}_d$ are the non-negative P, I and D controller fuzzification gain matrices [5]. For DP vessels, this control procedure can be illustrated in Fig. 9. The commanded forces and moments calculated by Eq. 1 are then distributed among the thrusters of the vessel. The actual forces generated by the propulsion system can be expressed as:

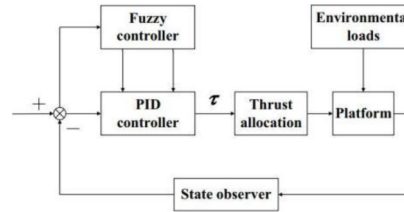


Figure 9—The schematic of DP controller with fuzzy rule-based controller

$$\tau = \mathbf{B}(\alpha)u \quad (2)$$

the vector $u \in \mathbb{R}^m$ contains the magnitude of the force produced by each individual actuator. The angle is the azimuth angle of the i -th actuator, defining the direction of the force produced in the body-fixed frame. The i -th column of the $3 \times m$ matrix $\mathbf{B}(\alpha)$ is given by:

$$B_i(\alpha_i) = \begin{pmatrix} \cos \alpha_i \\ \sin \alpha_i \\ l_{xi} \sin \alpha_i - l_{yi} \cos \alpha_i \end{pmatrix} \quad (3)$$

where (l_{xi}, l_{yi}) is the location of the i -th actuator in the body-fixed coordinate system. The thruster allocation at each time-step can be formulated as a quadratic programming problem [6].

$$\begin{aligned} \min J(\Delta\alpha, \mathbf{u}, \mathbf{s}) &= \mathbf{u}^T \mathbf{W} \mathbf{u} + \mathbf{s}^T \mathbf{Q} \mathbf{s} + \Delta\alpha^T \Omega \Delta\alpha \\ \mathbf{s} + \mathbf{B}(\alpha_0) \mathbf{u} + \frac{\partial}{\partial \alpha} (\mathbf{B}(\alpha) \mathbf{u}) \Big|_{\alpha=\alpha_0, \mathbf{u}=\mathbf{u}_0} \Delta\alpha &= \tau \\ \mathbf{u}_{\min} &\leq \mathbf{u} \leq \mathbf{u}_{\max} \\ \Delta\alpha_{\min} &\leq \Delta\alpha \leq \Delta\alpha_{\max} \end{aligned} \quad (4)$$

where $\mathbf{W}$, $\mathbf{Q}$ and Ω are the weighting matrix, $\Delta\alpha$ is the angle difference between this sample and last sample, $\mathbf{s}$ is the slack variable to make sure that the optimization problem always has a solution. The maximum and minimum forces of the actuators are specified through the constraints in Eq. 4. The mechanical constraints of the azimuth actuators are expressed in Eq. 4. The force produced by the propeller-rudder pair can be determined by its propeller revolution rate and rudder angle in the grid position, which can be treated as an azimuth thruster mathematically.

Results and discussion

Model calibration results

In present experiments, real-time data acquisition was performed to record the six degree of freedom (6DOF) motion time histories of the vessel. The spectral and correlation spectral analysis were applied to these motion responses to derive the corresponding Response Amplitude Operators (RAOs) over a period range of 4s-20s. The RAO results from the model tests were compared to the computed RAOs from diffraction theory as illustrated in the Fig. 10 and 11 for 0deg. and 90deg headings during ZCL stage. The vessel's motion response under white noise wave excitation generally showed good agreement with the results from diffraction analysis. Similar trends were observed across other floatover stages considered in the model tests, confirming consistency between experimental and numerical RAOs. These results validate the accuracy of the physical models and confirm their suitability for the intended experimental objectives.

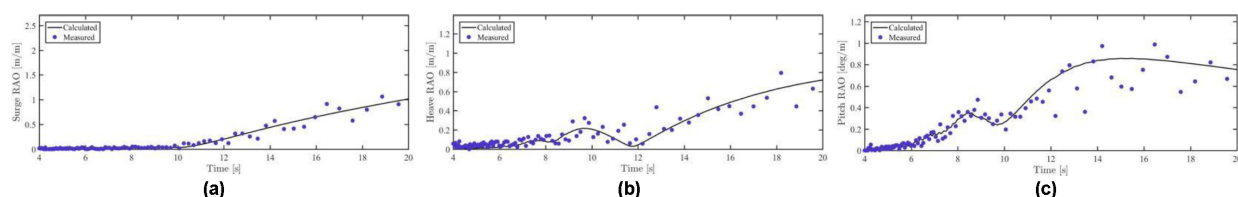


Figure 10—The RAO comparison between experimental data and numerical analysis for 0deg heading:(a) Surge; (b) Heave; (c) Pitch

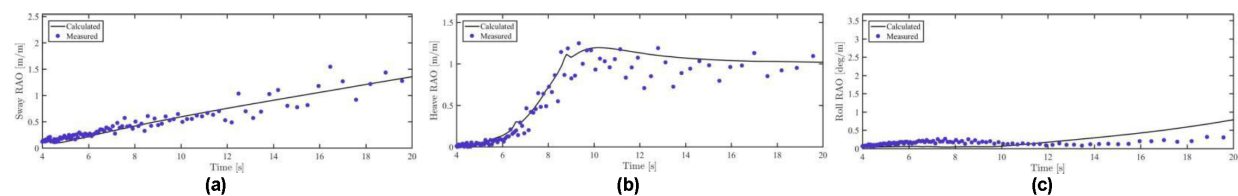


Figure 11—The RAO comparison between experimental data and numerical analysis for 90deg heading :(a) Sway; (b) Heave; (c) Roll

DP calibration results. Table 3 summarizes the full-scale results of the DP standby tests, while Fig. 12 illustrates the vessel trajectories under various wave directions. Overall, the physical DP control system demonstrated excellent station-keeping performance in the sea states described in the previous section. Maximum yaw motions were around 5° , with the largest deviation of 4.93° observed at 0° wave heading. For surge and sway motions, the maximum deviation reached 12.21 m in full-scale, equivalent to 30 cm in model scale. These larger deviations were primarily caused by instantaneous wave loads, although most displacements remained within ± 5 m. Sway motions were notably observed at 0° , and surge motions at 90° . At 0° , this behavior was mainly due to high-frequency switching of yaw moment direction, requiring the four-channel thrusters to alternate between forward and reverse thrust. Due to inherent thruster hysteresis, this switching caused temporary thrust insufficiency, resulting in increased sway motion. Similarly, at 90° , frequent thrust reversals of Thruster No. 6 led to insufficient positioning force, again due to hysteresis effects. In conclusion, the DP control system used in this study has proven capable of maintaining stable vessel positioning under various wave conditions, making it well-suited for simulating physical docking and undocking operations during topside floatover installation.

Table 5—Summary results of the DP Standby results

Heading	Mode	Max. Absolute Value
0deg	Surge	7.67 m
	Sway	6.00 m
	Yaw	4.93deg
45deg	Surge	6.18m
	Sway	5.83m
	Yaw	4.83deg
90deg	Surge	12.21m
	Sway	11.38m
	Yaw	4.69deg

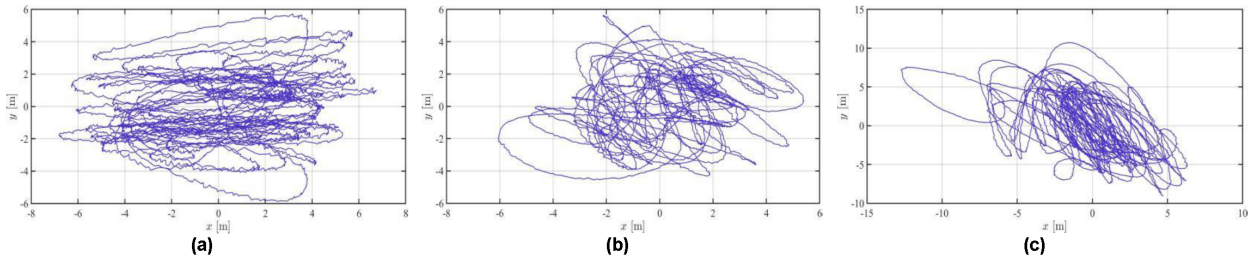


Figure 12—The trajectory of the vessel in different wave headings:(a) 0deg; (b) 45deg; (c) 90deg

Comparative analysis

DP docking and undocking tests

In the present study, the MOSES software is employed to perform hydrodynamic analyses based on potential flow theory, enabling comparison with scaled physical model tests. Figure 13 presents an overview of the docking tests, while Figure 14 illustrates a representative time history of a docking event from the model tests. Figures 13 and 14 compare the sway fender loads obtained from numerical simulations and scaled physical tests for both docking and undocking scenarios.

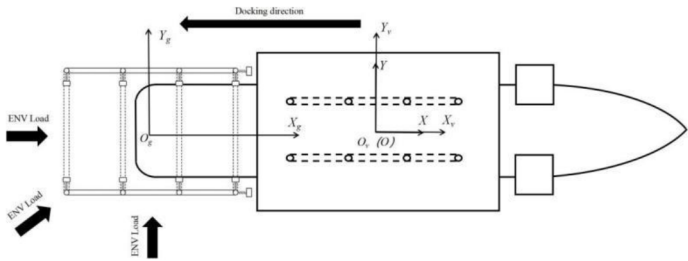


Figure 13—The sketch of docking tests

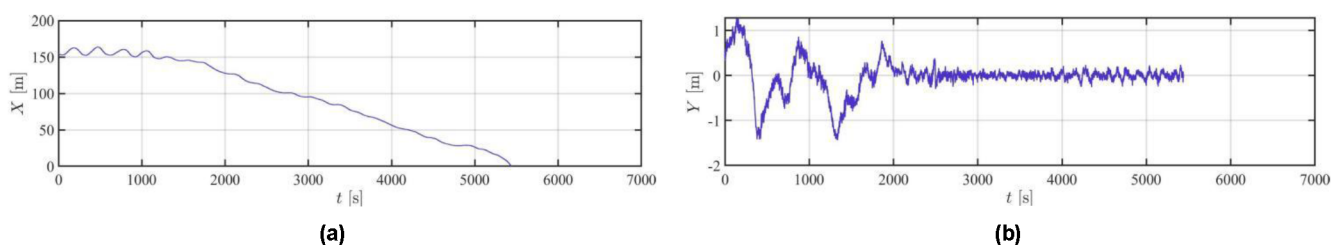


Figure 14—The time history of the docking tests, 0deg:(a) Surge motions (b) Sway motions

It is observed that the sway fender loads from the physical model tests are generally higher than those predicted by numerical simulations, particularly for the 0° heading during docking. The maximum sway fender load in the Z-direction recorded during the docking test is 126.6 MT on fender A4, whereas the numerical simulation predicts 0 MT. This discrepancy can be attributed to fundamental differences in the simulation methodologies. In the physical model tests, a scaled dynamic positioning (DP) system equipped with a fuzzy-PID controller is implemented to approximate real-world behavior. During docking and undocking, the vessel-coupled with the DP system intermittently contacts the sway fenders due to environmental disturbances and control limitations. In contrast, the numerical simulations assume ideal station-keeping using five mooring lines, which maintain the vessel's position greater precision and stability. Moreover, for the 0° heading case, numerical simulations yield zero transverse loads due to symmetry assumptions. However, in practice, diffraction effects from surrounding structures introduce additional forces, resulting in higher sway fender loads in the physical tests. Therefore, it is theoretically expected that scaled physical simulations will predict higher sway fender loads than numerical simulations.

Overall, the maximum sway fender load recorded during the model tests for both docking and undocking was approximately 360 MT, compared to around 320 MT obtained from numerical simulations. This close agreement indicates that the numerical model provides a reasonably accurate representation of the physical behavior, and both values remain within the design limits of the sway fender and the jacket structure.

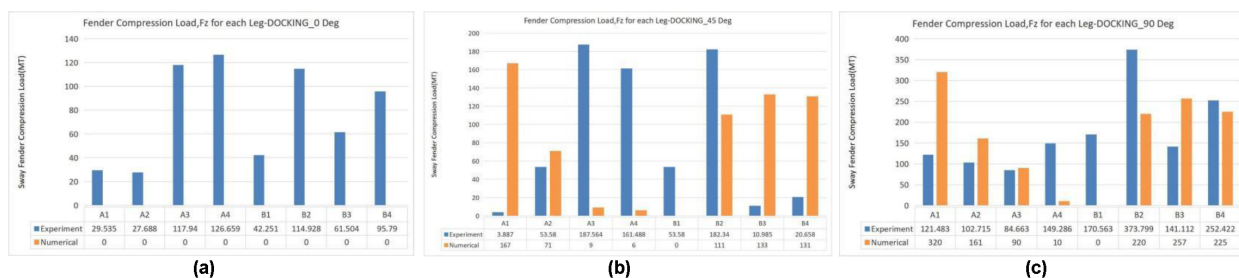


Figure 15—Maximum sway fender loads in docking tests:(a) 0deg; (b) 45 deg; (c) 90 deg

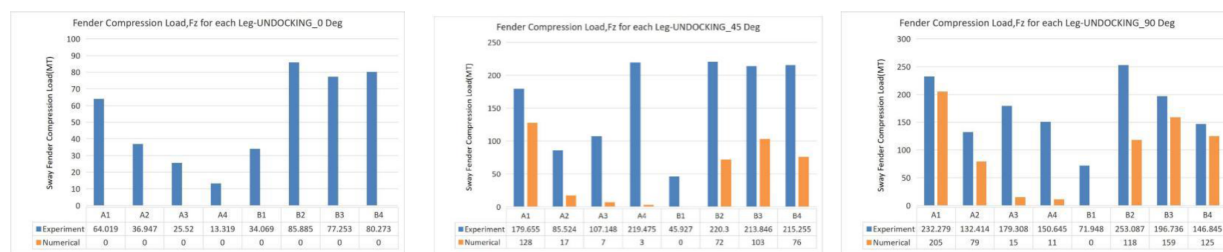


Figure 16—Maximum sway fender loads in undocking tests:(a) 0deg; (b) 45 deg; (c) 90 deg

Mating tests. Mating tests, also referred to as load transfer tests, simulate the gradual transfer of topside weight from the semi-submersible vessel onto the jacket structure. This process involves multiple stages of load transfer. In the MOSES software, these stages namely entrance, 0%, 30%, 70%, 100%, and exit are

individually simulated using a horizontal mooring system based on three-dimensional potential flow theory. A comparable setup is adopted in the scaled physical simulations, where each stage is tested separately.

Among all mating stages, the 70%LT stage was identified as the most critical in terms of dynamic vertical loads on both the LMUs and DSUs. This stage is emphasized and analyzed in detail in this paper, as illustrated in Figure 17. For the 70%LT stage, the maximum LMU loads recorded are: $F_x = 127$ MT at 45° , $F_y = 98$ MT at 90° , and $F_z = 3728$ MT at 0° . The maximum DSU loads are: $F_x = 148$ MT at 45° , $F_y = 93$ MT at 90° , and $F_z = 1683$ MT at 90° . The vertical LMU and DSU loads from the physical tests show good agreement with the numerical results. These vertical loads are relatively large and prioritized in the model tests.

However, discrepancies are observed in the longitudinal and transverse LMU and DSU loads between the experimental and numerical results, except in certain cases. Several factors may contribute to these differences:

1. In numerical simulations, diffraction waves generated by the jacket and in-water equipment are not considered. These waves, when coupled with incident waves, can amplify vessel motion responses, resulting in increased horizontal loads.
2. Given the magnitude of vertical loads, the load sensors used for LMU and DSU have a wide measurement range, making them highly sensitive. At model scale, even minor contact can produce disproportionately large readings.
3. Although most horizontal LMU and DSU stiffness values are calibrated, deviations exceeding 5% in some cases may contribute to the observed discrepancies.
4. The measured loads include not only horizontal and vertical components but also coupling effects from other directions, self-weight, friction, and other interactions.
5. Numerical simulations assume perfect symmetry in wave loading for 0° and 90° headings, based on symmetric panel modeling. In contrast, achieving such symmetry in physical tests is challenging, and environmental disturbances can significantly affect load measurements.
6. Load sensors for LMU, DSU, and fenders are subject to environmental noise. Although high-frequency filtering is applied during testing, complete elimination of noise is difficult, potentially leading to inflated load readings.

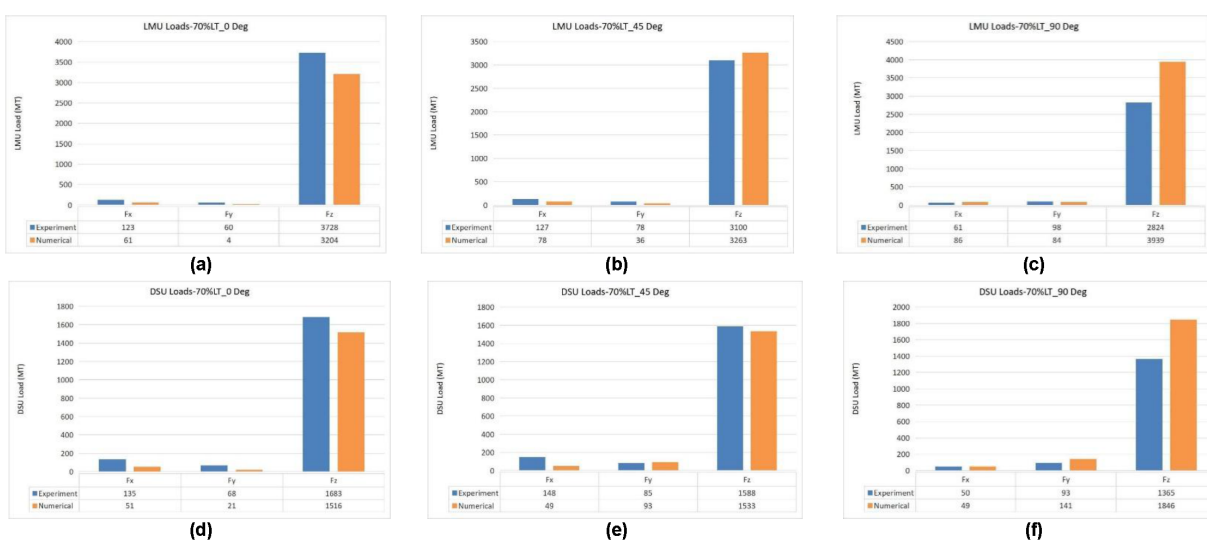


Figure 17—Maximum loads in 70%LT mating tests:(a) LMU 0deg; (b) LMU 45deg; (c) LMU 90deg; (d) DSU 0deg;(e) DSU 45deg;(f) DSU 90deg;

Conclusion

This paper provides a detailed introduction to the application of scaled physical simulation in mega offshore floatover installation. By integrating experimental data with numerical analyses performed using MOSES software, the accuracy of numerical predictions is validated, providing valuable support for engineering design decisions. The following key conclusions are drawn:

1. Motion response amplitude operators (RAOs) were determined using both white noise wave tests and diffraction analysis. Except for relatively high peak roll motions in beam sea conditions, good agreement was observed between the two methods across all stages. Deviations in peak roll RAOs, particularly when the wave period approaches the natural period, may be attributed to overestimation of viscous roll damping in numerical models. Additionally, the elevated wave height in white noise tests may introduce nonlinear effects, contributing to nonlinear roll responses.
2. During docking, the maximum absolute sway fender load recorded was 373.8 MT in beam sea, while during undocking, it was 253.1 MT. In physical tests, vessel movement was controlled via a fuzzy-PID-based dynamic positioning (DP) system, whereas numerical simulations employed five mooring lines for station keeping. The physical tests revealed intermittent contact between the vessel and sway fenders, reflecting realistic operational behaviour. In contrast, the numerical model maintained ideal positioning. This highlights the limitations of DP systems under high sea states, where station keeping performance may fall short of mooring-based systems.
3. For the critical 70% load transfer (70%LT) stage, the measured maximum LMU vertical load was 3728 MT and DSU vertical load was 1683 MT. Corresponding numerical predictions were 3929 MT and 1846 MT, respectively. In the horizontal direction, experimental LMU and DSU loads were 127 MT and 148 MT, compared to 86 MT and 141 MT from numerical simulations. These results indicate a reasonably good agreement between physical and numerical simulations.
4. Key Achievements from Scaled Physical Simulation include: a) The numerical model is validated for mating load predictions; b) Significant discrepancies were observed in docking and undocking simulations due to differing methodologies. Scaled physical simulations better reflect actual conditions, confirming that sway fender loads should not be zero at 0° and 45° headings; c) The scaled physical simulation supports the feasibility of DP-based floatover installation. Due to scale effects, real operations allow more time to respond to external disturbances. Moreover, while the model tests were conducted under upper-limit sea states, actual installations are planned during calm weather windows, further enhancing operation.

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